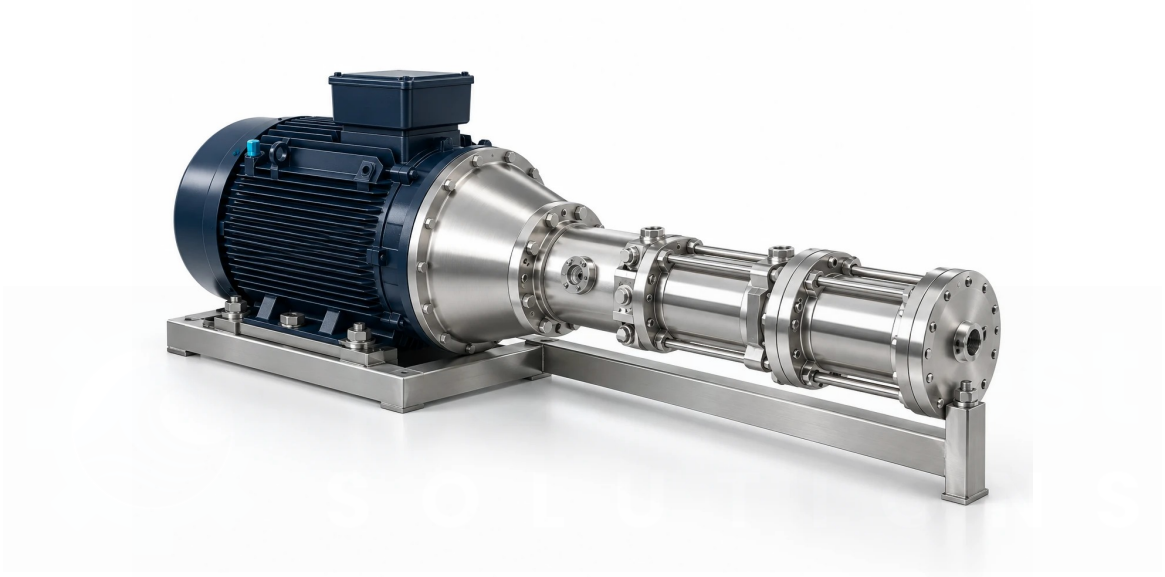
POWER CENTRE REVERSE OSMOSIS / PRODUCT INFORMATION

PC1000: feed pumping and brine energy recovery

Compare the PC1000 configuration at the required RO feed flow, pressure and recovery. Check feed and brine paths, motor input and any booster duty together; reference curves do not guarantee membrane-system production.

Two high-pressure axial piston pumps and an integrated energy recovery motor share one drive assembly. Built for larger seawater RO systems, with production up to 1,340 m³/day (353,991 US gal/day).



Recreated illustration of the PC1000 Power Centre Plus: electric motor, dual high-pressure pumps, integrated energy recovery motor and mounting frame

Key operating parameters

Parameter	Published value
Configuration	Dual axial piston pumps + integrated energy recovery motor
Housing, shaft and pistons	Duplex stainless steel
Operating medium	Sea / salt water
Maximum production	Up to 1,340 m³/day (353,991 US gal/day)
Operating pressure	30-85 bar (435-1,233 psi) listed; confirm upper limit

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Parameter	Published value
Configuration	Dual axial piston pumps + integrated energy recovery motor
Housing, shaft and pistons	Duplex stainless steel
Operating medium	Sea / salt water
Maximum production	Up to 1,340 m³/day (353,991 US gal/day)
Operating pressure	30-85 bar (435-1,233 psi) listed; confirm upper limit
Minimum inlet pressure	3.5 bar (50.8 psi) at pressure switch port
Maximum operating temperature	50 °C (122 °F)
Shaft speed	500-1,500 rpm
Mounting	Horizontal; support rear of unit
Drive connection	Splined shaft and flexible gear coupling
Power rating	Duty dependent - see performance curves
Filtration requirement	Confirm micron rating and water-quality limits

PC1000 connections · Technical reference p. 9

Port	Function	Connection
A	High-pressure out - pump 1	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
B	Low-pressure in - pump 1	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
C	Low-pressure in - pump 2	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
D	High-pressure out - pump 2	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
E	Case drain - pump 1	1½" SAE 4-bolt flange, 300 psi (20.7 bar (300 psi)) class - Confirm
F	Case drain - pump 2	1½" SAE 4-bolt flange, 300 psi (20.7 bar (300 psi)) class - Confirm
G	Case drain - energy recovery motor	1½" SAE 4-bolt flange, 300 psi (20.7 bar (300 psi)) class - Confirm
H	Low-pressure out - energy recovery motor	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
I	High-pressure in - energy recovery motor	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
J	Inlet pressure switch	G¼" BSPP female

PC1000 recovery & ordering references

Recovery	Order code
35%	P1000-35
40%	P1000-40
45%	P1000-45
50%	P1000-50

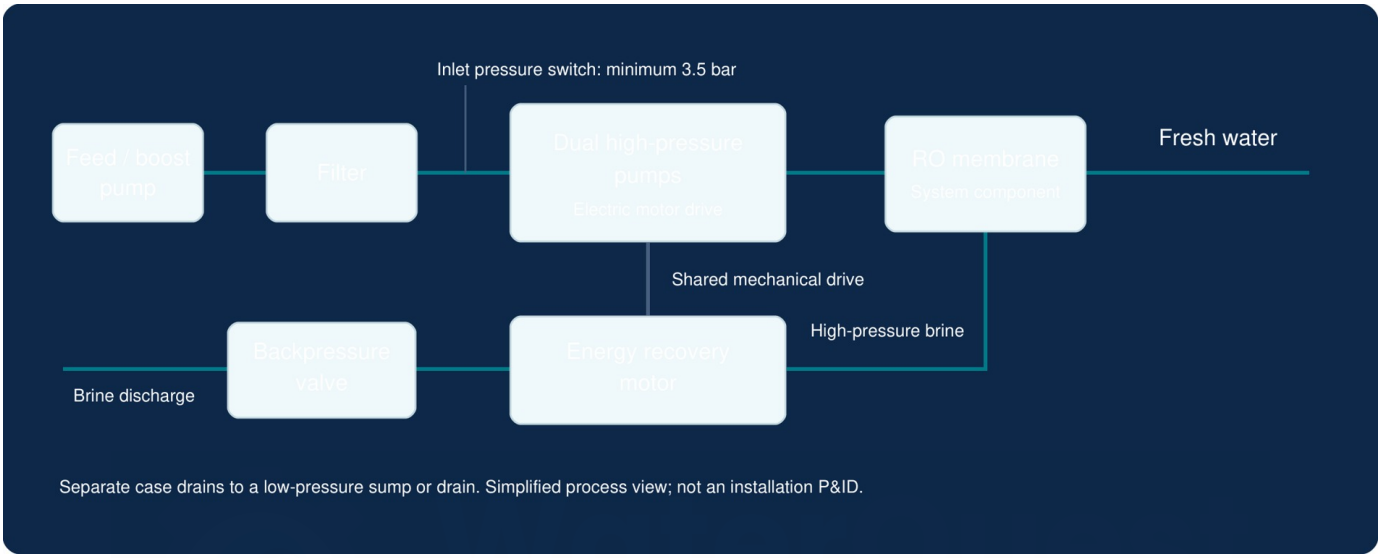
Selection and integration

Engineering checks Confirm before selection Pressure rating. The specification table lists 85 bar (1,233 psi) maximum; the circuit description states 80 bar (1,160 psi). Confirm the approved maximum for the selected unit. Case-drain flange rating. The connection table states 300 psi (20.7 bar (300 psi)) class, while the drawing notation requires clarification. Confirm the correct flange standard before ordering. Recovery options. Recovery is selected at the order stage. The listed codes cover 35%, 40%, 45% and 50%. Production and power. Confirm duty at your feed salinity, temperature, membrane selection, recovery and operating hours.

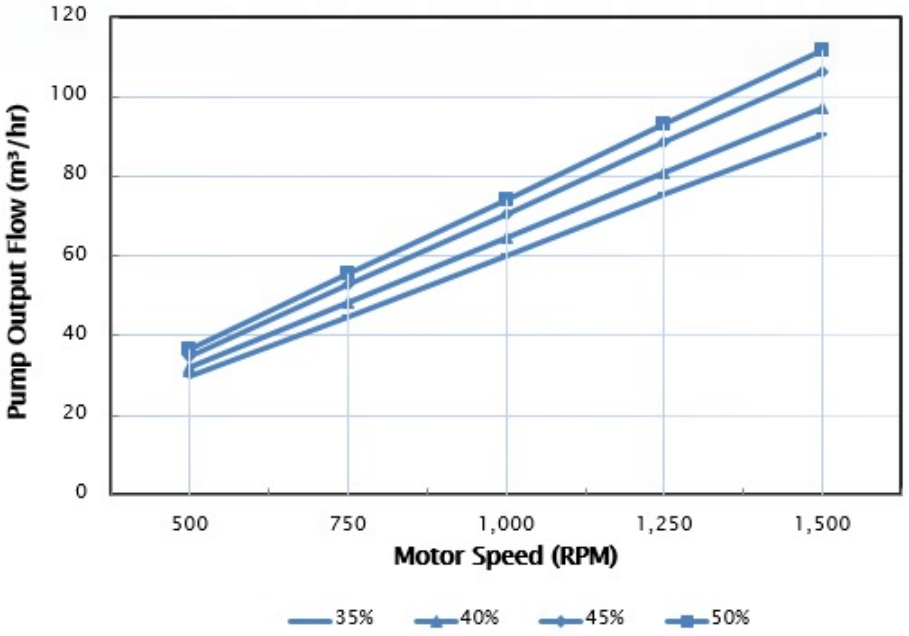
Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

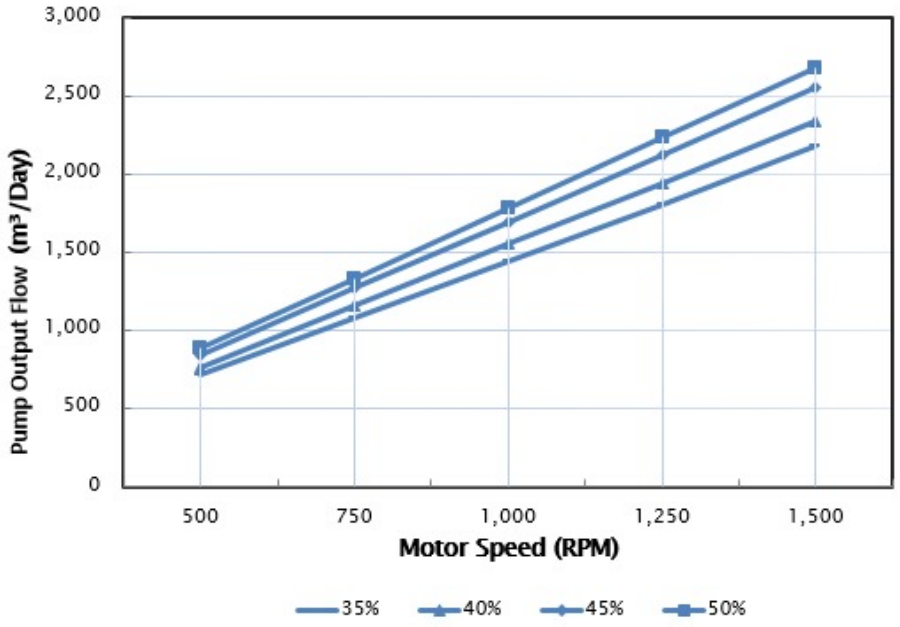
Process diagram: filtered feed enters the high-pressure pumps, the RO membrane produces fresh water, and reject brine returns through the energy recovery motor to assist the shared drive.



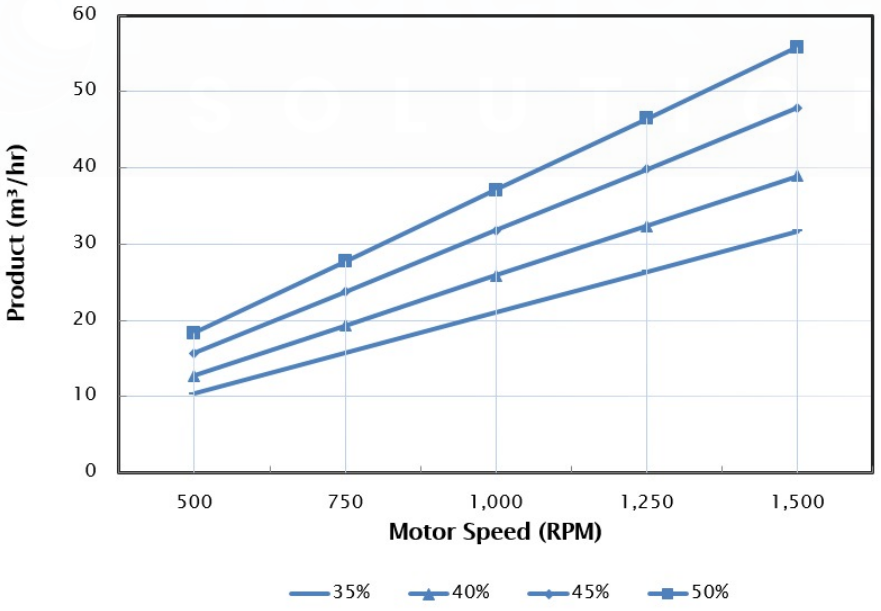
PC1000 pump output flow chart, m³/h. Flow varies with shaft speed and operating conditions.



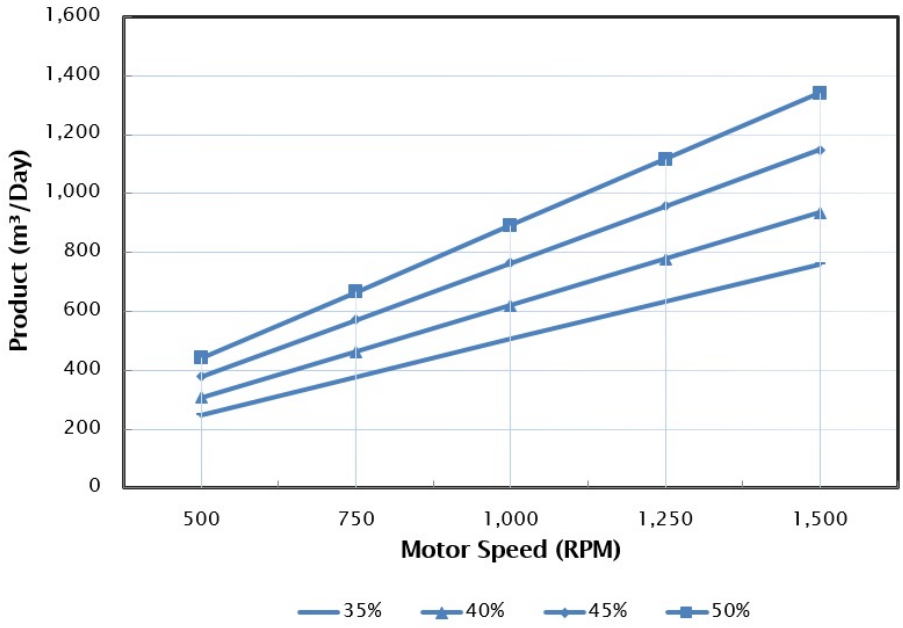
PC1000 pump output flow chart, m³/day. Flow varies with shaft speed and operating conditions.



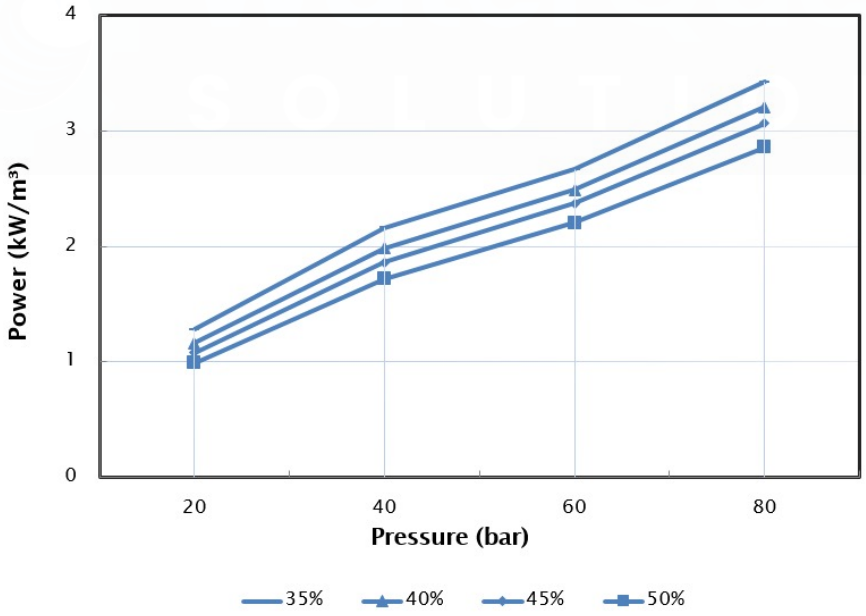
PC1000 freshwater production chart, m³/h. At 80 bar; recovery selected when ordering.



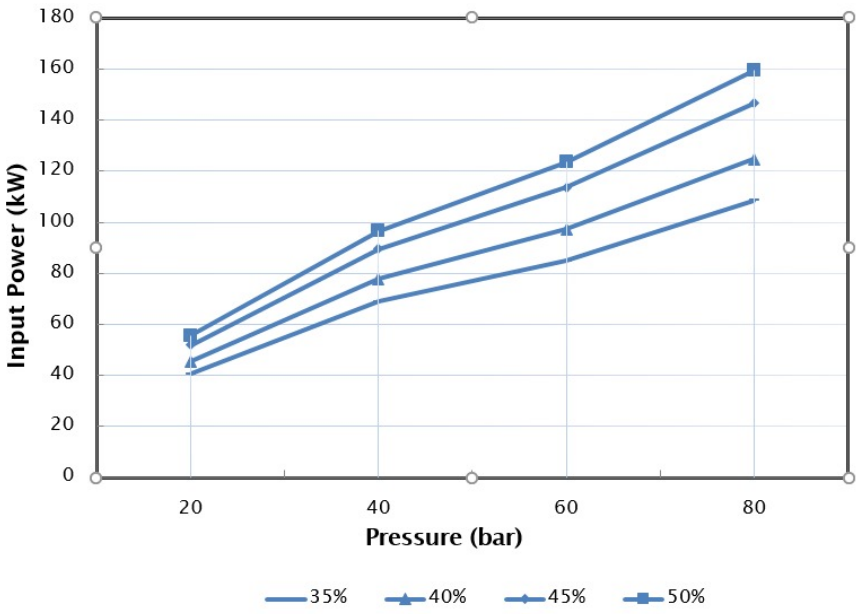
PC1000 freshwater production chart, m³/day. At 80 bar; recovery selected when ordering.



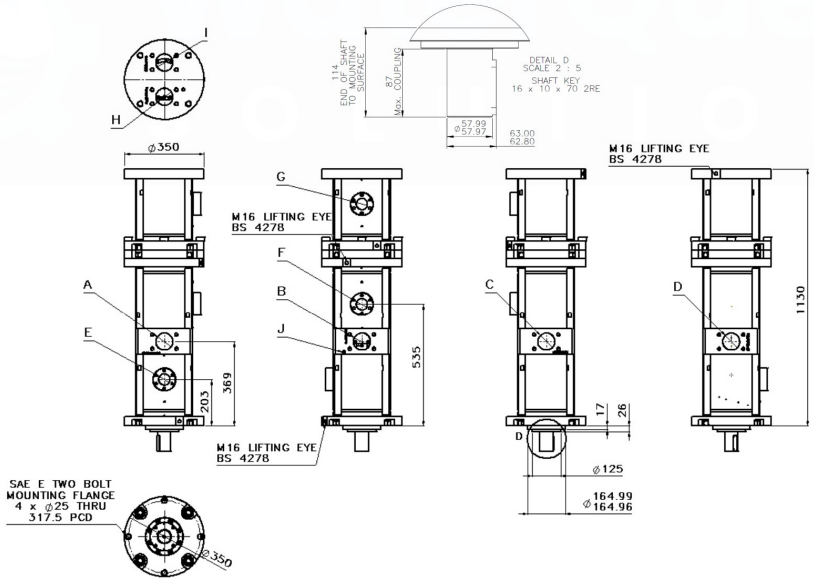
PC1000 specific energy chart, Chart units: kW/m³. Performance curves at 1,500 rpm.



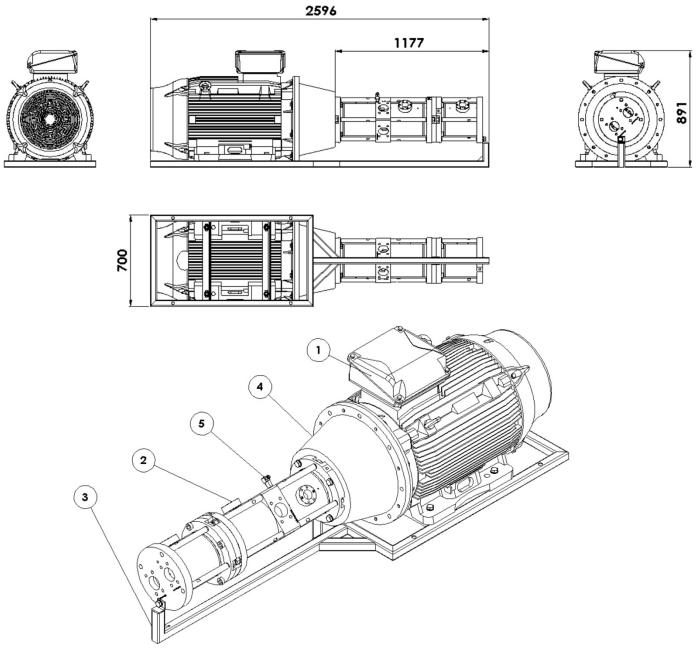
PC1000 input power chart, kW. Performance curves at 1,500 rpm.



PC1000 dimensioned unit drawing with port positions; open full-size for dimensions



PC1000 motor, coupling, Power Centre and mounting frame assembly drawing



PC1000 circuit schematic

